Stat 110 Strategic Practice 9 & Homework 9: Solutions

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Strategic Practice 9 Solutions

1. Beta and Gamma Distributions

Problem 1. Let $B \sim \text{Beta}(a, b)$.

(a) Let $W = 1 - B$. Since $g(t) = 1 - t$ is strictly decreasing with $|g'(t)| = 1$, the change-of-variables formula gives

$$f_W(w) = f_B(1-w) \cdot 1 = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}(1-w)^{a-1}w^{b-1}, \quad w \in (0,1).$$

This is the Beta(b, a) PDF, so $1 - B \sim \text{Beta}(b, a)$.

(b) Represent $B = \frac{X}{X+Y}$ where $X \sim \text{Gamma}(a, 1)$ and $Y \sim \text{Gamma}(b, 1)$ are independent (the bank-post office story). Then $1 - B = \frac{Y}{X+Y}$. By the same story with the roles of X and Y swapped, $\frac{Y}{X+Y} \sim \text{Beta}(b, a)$.

This connects naturally to the Beta-Binomial conjugate prior setup: if Beta(a, b) serves as a prior for success probability p (with a representing prior successes and b prior failures), then swapping the labels of success and failure turns p into $1 - p$, and the corresponding prior should be Beta(b, a), which is exactly $1 - B$.

Problem 2. $X \sim \text{Gamma}(a, \lambda)$ and $Y \sim \text{Gamma}(b, \lambda)$ independent, a and b positive integers.

(a) The convolution of the PDFs:

$$f_{X+Y}(t) = \int_0^t f_X(x) f_Y(t-x)\, dx = \int_0^t \frac{\lambda^a x^{a-1} e^{-\lambda x}}{\Gamma(a)} \cdot \frac{\lambda^b (t-x)^{b-1} e^{-\lambda(t-x)}}{\Gamma(b)}\, dx.$$

Pulling out the exponential and powers of λ,

$$= \frac{\lambda^{a+b} e^{-\lambda t}}{\Gamma(a)\Gamma(b)} \int_0^t x^{a-1}(t-x)^{b-1}\, dx.$$

Substituting $u = x/t$, the integral becomes $t^{a+b-1}B(a,b) = t^{a+b-1}\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$. So

$$f_{X+Y}(t) = \frac{\lambda^{a+b} e^{-\lambda t}}{\Gamma(a+b)} t^{a+b-1},$$

which is the Gamma$(a + b, \lambda)$ PDF.

(b) The MGF of Gamma(a, λ) is $\left(\frac{\lambda}{\lambda - t}\right)^a$ for $t < \lambda$. By independence, the MGF of $X + Y$ is

$$\left(\frac{\lambda}{\lambda - t}\right)^a \cdot \left(\frac{\lambda}{\lambda - t}\right)^b = \left(\frac{\lambda}{\lambda - t}\right)^{a+b},$$

which is the Gamma$(a + b, \lambda)$ MGF.

(c) Interpret X as the waiting time for the ath arrival in a Poisson process with rate λ. Since inter-arrival times are i.i.d. Expo(λ) and are independent of X, Y can be interpreted as the additional waiting

time for b more arrivals after the ath one. Then $X+Y$ is the waiting time until the $(a+b)$th arrival, so $X+Y \sim \text{Gamma}(a+b,\lambda)$.

Problem 3. Let $W = \frac{X}{X+Y}$ and $S = X+Y$. The bank-post office story establishes that $W \sim \text{Beta}(a,b)$ is independent of $S \sim \text{Gamma}(a+b,\lambda)$. Since $X/Y = W/(1-W)$ is a function of W alone, and $S = X+Y$ is a function of S alone, and W is independent of S, it follows that X/Y is independent of $X+Y$.

Problem 4. Let $X \sim \text{Gamma}(m/2,1/2)$ and $Y \sim \text{Gamma}(n/2,1/2)$ independently, and $V = \frac{X/m}{Y/n} = \frac{nX}{mY}$, so $X = \frac{mV\cdot Y}{n}$. Then

$$\frac{mV}{n+mV} = \frac{mV}{n+mV}.$$

Working directly: $mV/(n+mV) = X/(X+Y)$ after substituting $V = nX/(mY)$. By the bank-post office story, $X/(X+Y) \sim \text{Beta}(m/2,n/2)$.

2. Order Statistics

Problem 1. $U_1,\dots,U_n \overset{iid}{\sim} \text{Unif}(0,1)$, and $U_{(j)} \sim \text{Beta}(j,n-j+1)$. The PDF is

$$f_{U_{(j)}}(t) = \frac{n!}{(j-1)!(n-j)!}t^{j-1}(1-t)^{n-j}, \quad t \in (0,1).$$

For $T \sim \text{Beta}(a,b)$ with $a=j$ and $b=n-j+1$: $E(T) = \frac{a}{a+b} = \frac{j}{n+1}$.

For the variance, compute $E(T^2)$ by recognizing the integrand as a $\text{Beta}(a+2,b)$ kernel:

$$E(T^2) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\cdot\frac{\Gamma(a+2)\Gamma(b)}{\Gamma(a+b+2)} = \frac{a(a+1)}{(a+b)(a+b+1)}.$$

So $\text{Var}(T) = \frac{a(a+1)}{(a+b)(a+b+1)} - \frac{a^2}{(a+b)^2} = \frac{ab}{(a+b)^2(a+b+1)}$. With $a=j$, $b=n-j+1$,

$$\text{Var}(U_{(j)}) = \frac{j(n-j+1)}{(n+1)^2(n+2)}.$$

Problem 2. $X,Y \overset{iid}{\sim} \text{Expo}(\lambda)$, $M = \max(X,Y)$.

(a) The CDF of M is $F_M(x) = (1-e^{-\lambda x})^2$. For $X + \frac{1}{2}Y$, conditioning on $Y = t$:

$$F_{X+Y/2}(x) = \int_0^{2x}(1-e^{-\lambda(x-t/2)})\lambda e^{-\lambda t}\,dt = \int_0^{2x} e^{-\lambda t}\lambda\,dt - \int_0^{2x}\lambda e^{-\lambda x}e^{-\lambda t/2}\,dt.$$

Computing each integral and simplifying yields $(1-e^{-\lambda x})^2$, the same as $F_M(x)$.

(b) Write $M = \min(X,Y) + (M - \min(X,Y))$. The minimum $L = \min(X,Y) \sim \text{Expo}(2\lambda)$. At the moment L is realized, the remaining variable (whichever of X,Y was not the minimum) has residual time $\sim \text{Expo}(\lambda)$ by the memoryless property, independent of L. So $M - L \sim \text{Expo}(\lambda)$ independently of L. Now $\frac{1}{2}Y \sim \text{Expo}(2\lambda)$ and $X \sim \text{Expo}(\lambda)$ are independent, so $L + \text{Expo}(\lambda)$ and $\frac{1}{2}Y + X$ have the same joint structure and thus the same distribution.

Problem 3. To have $X_{(i)} \approx a$ and $X_{(j)} \approx b$ (for $a < b$), imagine sorting the n observations into five groups: the $i-1$ values below a, the one value near a, the $j-i-1$ values between a and b, the one value near b, and the $n-j$ values above b. The number of ways to assign the n observations to these groups (with sizes $i-1, 1, j-i-1, 1, n-j$) is the multinomial coefficient $\frac{n!}{(i-1)!\,1!\,(j-i-1)!\,1!\,(n-j)!}$. Each configuration contributes probability $F(a)^{i-1}f(a)\,(F(b)-F(a))^{j-i-1}f(b)\,(1-F(b))^{n-j}$ times the infinitesimal areas $da\,db$. So the joint PDF is

$$f_{X_{(i)},X_{(j)}}(a,b) = \frac{n!}{(i-1)!\,(j-i-1)!\,(n-j)!}F(a)^{i-1}(F(b)-F(a))^{j-i-1}(1-F(b))^{n-j}f(a)f(b),$$

for $a < b$.

3. Conditional Expectation

Problem 1. Let the two envelope values be $a < b$ (the case $a = b$ is trivial). We open our envelope and see x, then draw $T \sim \text{Expo}(1)$ and switch if $x < T$.

If we initially picked the envelope with value a: we switch (and win) whenever $T > a$, which has probability e^{-a}.

If we initially picked the envelope with value b: we switch (and lose) whenever $T > b$; otherwise we keep and win. The probability of winning is $P(T \le b) = 1 - e^{-b}$.

Each envelope is picked with probability $1/2$, so the overall success probability is

$$\frac{1}{2}e^{-a} + \frac{1}{2}(1 - e^{-b}) = \frac{1}{2} + \frac{1}{2}(e^{-a} - e^{-b}).$$

Since $a < b$, we have $e^{-a} > e^{-b}$, so $e^{-a} - e^{-b} > 0$, making this probability strictly greater than $1/2$.

Intuitively: if T happens to fall between a and b (i.e., $a < T < b$), the strategy works perfectly (switch when holding a, keep when holding b). If T falls outside the interval, the strategy reduces to random guessing at $1/2$ but never does strictly worse. The possibility of T landing in the interval provides a strict improvement.

Problem 2. Let $X \sim \text{Geom}(p)$ (number of failures before first success). Condition on whether the first trial is a success:

$$E(X) = E(X \mid \text{success}) \cdot p + E(X \mid \text{failure}) \cdot q = 0 \cdot p + (1 + E(X)) \cdot q,$$

since after a failure the process restarts. Solving: $E(X) = q + qE(X)$, so $E(X)(1-q) = q$, giving $E(X) = q/p$.

Problem 3. Let a_j be the expected number of additional days to complete the collection when j numbers are still missing. Condition on N, the count of new (previously missing) numbers drawn on the next day. Since 5 balls are drawn without replacement from 35 total, N follows a Hypergeometric distribution (tagging the j missing numbers):

$$P(N = n) = \frac{\binom{j}{n}\binom{35-j}{5-n}}{\binom{35}{5}}.$$

After that day, $j - N$ numbers remain missing, so

$$a_j = 1 + \sum_{n=0}^{5} a_{j-n} \cdot P(N = n), \quad j = 1, \ldots, 35,$$

with $a_0 = 0$ and $a_k = 0$ for $k < 0$ (no more numbers to collect if $j - n \le 0$).

Problem 4. The bid b is accepted when $b \ge \frac{2}{3}V$, i.e., $V \le \frac{3}{2}b$. For $b \ge 2/3$ the bid is always accepted but $E(V - b) = E(V) - b = 1/2 - b < 0$, so never bid that high. For $b < 2/3$,

$$E(\text{payoff}) = E(V - b \mid V \le \tfrac{3}{2}b) \cdot P(V \le \tfrac{3}{2}b).$$

Given $V \le \frac{3}{2}b$, the conditional distribution of V is $\text{Unif}(0, \frac{3}{2}b)$, so $E(V \mid V \le \frac{3}{2}b) = \frac{3}{4}b$. Also $P(V \le \frac{3}{2}b) = \frac{3}{2}b$. Therefore,

$$E(\text{payoff}) = \left(\frac{3}{4}b - b\right) \cdot \frac{3}{2}b = -\frac{1}{4}b \cdot \frac{3}{2}b = -\frac{3}{8}b^2 \le 0,$$

with equality only at $b = 0$. The optimal bid is $b = 0$: do not play. The key lesson is that accepting the bid is informative: it signals that V is small relative to b, so conditioning on acceptance dramatically lowers the expected prize value below the unconditional mean of $1/2$.

Homework 9 Solutions

1. Expectation of a Ratio

(a) Let $X = 1$ (constant) and $Y = 1$ or 3 each with probability 1/2. Then $E(X)/(E(X) + E(Y)) = 1/(1+2) = 1/3$, while $E(X/(X+Y)) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} = \frac{3}{8} \neq \frac{1}{3}$.

(b) Yes. When X, Y are i.i.d., by symmetry $E(X/(X+Y)) = E(Y/(X+Y))$. Adding and using linearity, $2E(X/(X+Y)) = E((X+Y)/(X+Y)) = 1$, so $E(X/(X+Y)) = 1/2$. On the other hand $E(X)/(E(X) + E(Y)) = E(X)/(2E(X)) = 1/2$.

(c) From the bank-post office story, $W = X/(X+Y)$ is independent of $S = X+Y$ when $X \sim \text{Gamma}(a, \lambda)$ and $Y \sim \text{Gamma}(b, \lambda)$. Since $X^c/(X+Y)^c = W^c$ is a function of W alone and $(X+Y)^c = S^c$ is a function of S alone, and $W \perp S$, we get $W^c \perp S^c$. For independent random variables,

$$E(W^c \cdot S^c) = E(W^c)E(S^c),$$

and $W^c \cdot S^c = (W \cdot S)^c = X^c$, so $E(X^c) = E(W^c)E(S^c) = E\left(\frac{X^c}{(X+Y)^c}\right) E((X+Y)^c)$, which gives the claimed identity.

2. The Gumbel Distribution

(a) Let $G = -\log X$ with $X \sim \text{Expo}(1)$. The CDF of G is

$$P(G \leq t) = P(-\log X \leq t) = P(X \geq e^{-t}) = e^{-e^{-t}}, \quad t \in \mathbb{R}.$$

(b) The CDF of $M_n - \log n$ is

$$P(M_n - \log n \leq t) = P(X_1 \leq t + \log n)^n = (1 - e^{-(t+\log n)})^n = \left(1 - \frac{e^{-t}}{n}\right)^n.$$

Using $(1 + x/n)^n \to e^x$ as $n \to \infty$ (here $x = -e^{-t}$), this converges to $e^{-e^{-t}}$, the Gumbel CDF.

3. Negative Binomial Converges to Gamma

(a) Let $X \sim \text{NBin}(r, p)$ (number of failures before r successes) and $q = 1 - p$. The MGF of a single Geom(p) r.v. is $p/(1 - qe^t)$ for $qe^t < 1$, so the MGF of X is $\left(\frac{p}{1-qe^t}\right)^r$. The MGF of the scaled variable $Y_p = \frac{p}{q}X$ is obtained by substituting tp/q for t:

$$M_{Y_p}(t) = \left(\frac{p}{1 - qe^{tp/q}}\right)^r.$$

As $p \to 0$, use the approximation $e^{tp/q} \approx 1 + tp/q + O(p^2/q^2)$, so $qe^{tp/q} \approx q + tp + O(p^2) \approx 1 - p + tp + O(p^2)$. Then $1 - qe^{tp/q} \approx p - tp + O(p^2) = p(1-t) + O(p^2)$, and

$$M_{Y_p}(t) \approx \left(\frac{p}{p(1-t)}\right)^r = \frac{1}{(1-t)^r},$$

which is the Gamma$(r, 1)$ MGF (valid for $t < 1$). By the MGF convergence theorem, $Y_p \to \text{Gamma}(r, 1)$ in distribution as $p \to 0$.

(b) The Negative Binomial counts discrete failures before r successes, and the Gamma is its continuous counterpart: both measure the "waiting time" (or total effort) to accumulate r events at a given rate. As $p \to 0$ the trials become very frequent and each has a tiny success probability, making the total count large; the scaling $p/q \approx p$ converts discrete failure counts into a continuous waiting time. Just as the Geometric (with $r = 1$) converges to the Exponential under the same scaling, the Negative Binomial converges to the Gamma.

4. Maximum of Discrete vs. Continuous

(a) For continuous i.i.d. X, Y: $P(M = x)$ comes purely from the event that exactly one equals x while the other is less, since ties have probability zero. In the discrete case, $P(X = Y = x) > 0$, so the formula must account for the tie event separately, breaking the simple $2F(x)f(x)$ form.

(b) $X, Y \overset{iid}{\sim} \text{Bern}(1/2)$. The joint PMF of (M, L):

$$P(M = 0, L = 0) = P(X = 0, Y = 0) = \tfrac{1}{4},$$
$$P(M = 1, L = 0) = P(X = 1, Y = 0) + P(X = 0, Y = 1) = \tfrac{1}{2},$$
$$P(M = 1, L = 1) = P(X = 1, Y = 1) = \tfrac{1}{4},$$
$$P(M = 0, L = 1) = 0 \text{ (impossible since } M \geq L).$$

Marginal PMF of M: $P(M = 0) = 1/4$, $P(M = 1) = 3/4$, so $M \sim \text{Bern}(3/4)$.

Marginal PMF of L: $P(L = 0) = 3/4$, $P(L = 1) = 1/4$, so $L \sim \text{Bern}(1/4)$.

5. Beta-Binomial CDF Identity

Let $U_1, \ldots, U_n \overset{iid}{\sim} \text{Unif}(0, 1)$ and define a trial as "successful" if $U_k \leq p$. Each trial independently succeeds with probability p, so the total number of successes $X \sim \text{Bin}(n, p)$. The jth order statistic $U_{(j)}$ is the jth smallest among $U_1, \ldots, U_n$, and $U_{(j)} \sim \text{Beta}(j, n - j + 1)$.

The key observation is: $\{X \geq j\} = \{U_{(j)} \leq p\}$. This is because $X \geq j$ means at least j of the U_k's fall in $[0, p]$, which is exactly the condition that the jth smallest U_k is at most p. Therefore,

$$P(X \geq j) = P(U_{(j)} \leq p) = P(B \leq p),$$

where $B \sim \text{Beta}(j, n - j + 1)$.

6. Expected Flips Until a Pattern

(a) Waiting for HT: first wait for an H (expected $1/p$ flips), then wait for the first T after that (expected $1/q$ additional flips, since each flip after the first H is T with probability q). These phases are independent, so $E(\text{flips until HT}) = \frac{1}{p} + \frac{1}{q}$.

(b) Waiting for HH: let $\mu = E(\text{flips})$. Condition on the first flip:

$$\mu = p \cdot E(\text{flips} \mid \text{first is H}) + q \cdot (1 + \mu),$$

since after a T the process restarts. Given the first flip is H, condition on the second:

$$E(\text{flips} \mid \text{first is H}) = p \cdot 2 + q \cdot (2 + \mu).$$

Substituting: $\mu = p(2p + q(2 + \mu)) + q(1 + \mu)$. Expanding and solving for μ:

$$\mu = 2p + pq(2 + \mu) + q + q\mu \implies \mu p^2 = p^2 \cdot 2 + pq + q = \frac{p + q}{p} + 1 = \frac{1}{p} + 1,$$

giving $\mu = \frac{1}{p} + \frac{1}{p^2}$.

(c) Now $p \sim \text{Beta}(a, b)$ is unknown. By iterated expectation, $E(\text{flips until HT}) = E\left[\frac{1}{p} + \frac{1}{q}\right] = E(1/p) + E(1/q)$.

Using the Beta PDF and the identity $\Gamma(a - 1) = (a - 2)!$ etc.:

$$E\left(\frac{1}{p}\right) = \frac{a + b - 1}{a - 1}, \qquad E\left(\frac{1}{q}\right) = \frac{a + b - 1}{b - 1},$$

so the expected flips until HT is $\frac{a+b-1}{a-1} + \frac{a+b-1}{b-1}$.

Similarly, $E(\text{flips until HH}) = E(1/p) + E(1/p^2)$.

$$E\left(\frac{1}{p^2}\right) = \frac{(a + b - 1)(a + b - 2)}{(a - 1)(a - 2)},$$

so the expected flips until HH is $\frac{a+b-1}{a-1} + \frac{(a+b-1)(a+b-2)}{(a-1)(a-2)}$.

7. Roommate Pairs in Stat 110

Let I_j indicate that both members of roommate pair j take Stat 110, so $X = \sum_{j=1}^{n} I_j$. Each student independently takes Stat 110 with probability p, so $P(I_j = 1) = p^2$ and

$$E(X) = np^2.$$

For $E(X \mid N)$: given that N of the $2n$ students enrolled, we condition on N without knowing which pairs they form. For any fixed pair, the probability that student 1 of the pair is in the group of N is $\frac{N}{2n}$, and given that, the probability the partner also is in the group is $\frac{N-1}{2n-1}$. So

$$E(I_j \mid N) = \frac{N}{2n} \cdot \frac{N-1}{2n-1} = \frac{N(N-1)}{2n(2n-1)}.$$

By linearity,

$$E(X \mid N) = n \cdot \frac{N(N-1)}{2n(2n-1)} = \frac{N(N-1)}{2(2n-1)}.$$